

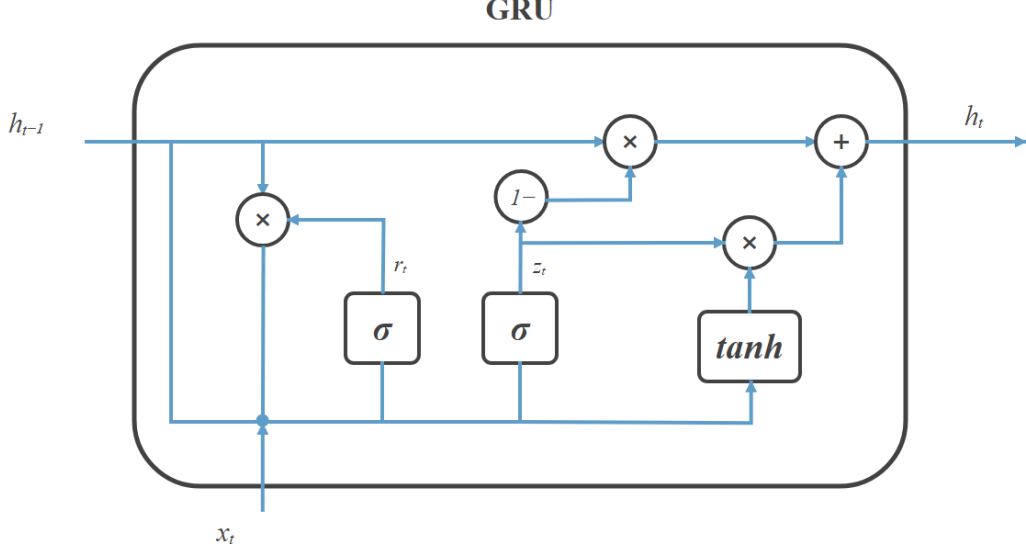


Figure 3.4: Detailed data flow through a GRU cell corresponding to Equation (3.14). The reset gate controls the historical contribution to the candidate state. In the convention used here, $\mathbf{z}_t$ weights the newly computed candidate and $1 - \mathbf{z}_t$ weights the previous hidden state. Unlike the LSTM, the GRU propagates only one recurrent state.

Equation (3.14) and Figure 3.4 use the candidate-weight convention and the reset-before formulation. The update gate therefore weights $\tilde{\mathbf{h}}_t$ directly, and $\mathbf{r}_t$ is applied to $\mathbf{h}_{t-1}$ before the recurrent matrix multiplication. An equally common notation lets the update gate weight the previous state instead. With the complementary gate $\mathbf{z}'_t = 1 - \mathbf{z}_t$, the same update can be written as

$$\mathbf{h}_t = (1 - \mathbf{z}'_t) \odot \tilde{\mathbf{h}}_t + \mathbf{z}'_t \odot \mathbf{h}_{t-1}. \quad (3.15)$$

Implementations may also apply the reset gate either before or after the recurrent affine transformation. These alternatives leave the tensor dimensions unchanged, but the equations, trained parameters, and recurrent states must be interpreted using the same convention.

For input dimension D and hidden dimension H , the update gate, reset gate, and candidate each require one input and one recurrent transformation. With one combined bias per transformation, the parameter count is

$$N_{\text{param}}^{\text{GRU}} = 3H(D + H + 1). \quad (3.16)$$